

A Data-Driven AI Approach for Predicting and Enhancing Individual Environmental Sustainability Scores

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Abstract- Sustainability in the environment is becoming increasingly a key factor in the health of society and our planet.^[1] Despite many guidelines and calculators, there exists a requirement for a comprehensive data-based approach combining diverse parameters of lifestyles into a singular, quantifiable score.^[3] In this paper, we introduce an AI model that forecasts a person's sustainability score based on various inputs such as monthly electricity usage, modes of transportation, car ownership, types of fuel, frequency of flight travel, recycling behavior, and electronics buying habits. Our system employs a combination of regression methods and feature engineering to transform every parameter into a standardized CO₂ equivalent or percentage effect. We then aggregate these values through a weighted scoring algorithm to generate a single sustainability index for each user.

Keywords- Sustainability · CO₂ · AI Model · Data-Driven · Lifestyle · Environmental Impact · Regression · Feature Engineering · Transportation · Renewable Energy · Recycling Habits · Electronics Consumption · Carbon Footprint · Household Analysis · Green Technology.

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1 INTRODUCTION

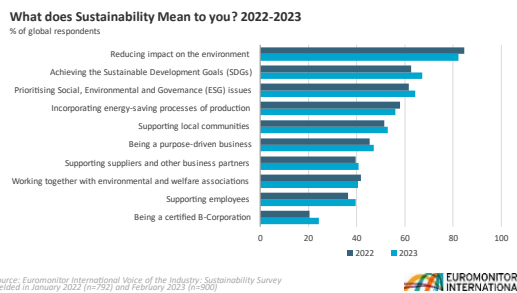
Redefining sustainability puts it right in the middle both in scientific debate and public discourse, as concern for climate change and depletion of resources is becoming increasingly real, creating the environmental crisis of today's modern lifestyle.^[2]

Governments, corporations, and individuals are all looking for ways to minimize carbon footprints and waste, as well as shift toward renewables.^[2] However, it has not yet devolved into a reliable tool for measuring the "cumulative" impact of different lifestyle choices. The problem with most conventional carbon footprint calculators is that they are often not granular or flexible enough to capture the minutiae of personal habits. This paper is hence intended to address this identified niche by proposing an AI model for evaluating an individual's sustainability score based on ten input factors from monthly kilowatt-hour usage, vehicle ownership, flight frequency, and recycling habits. These will all be translated into standardized parameters like CO₂ equivalent or percentage impact to aggregate the outputs into a single figure: an environmental snapshot of a given individual. The model we ultimately want to build is one that is scalable and transparent yet easy for one to interpret when directing them into behavior transformation for greater sustainability. That's where we believe the power of data will transform into data-driven cause-action and the driving force behind this acceleration of movement on a more sustainable future.

2 RELATED WORKS

Within the more extensive precincts of sustainability research, various studies have worked into methods of measuring the impacts, whether as individuals or collectively, on the environment. Carbon footprint calculators typically depend on generalized national averages or on some simplifications of household consumption patterns.^[3] More recently, AI and machine learning techniques have been applied for optimizing energy usage in smart grids, predicting traffic congestion, and enhancing waste management systems.^[11] *Life Cycle Assessment (LCA)* has been used so far in industrial contexts for the evaluation of overall environmental costs of products and services.^[8] As important as these contributions are, a nagging need is overwhelmingly apparent for a general individual-level model integrating numerous lifestyle variables into one easy-to-use score. *Our approach would go beyond that model with a stronger adaptive model that could be tied to areas, household sizes, and behaviors.*

SURVEY



Graph 1: "Survey on What Sustainability Means to Humans"

for instance, solar panel integration). The Transport Module will calculate emissions based on a user's choice of travel methods and fuel. The charts from the sustainability survey show the visual data

foundation upon which our project has built its approach to using individuals for modeling sustainability. In our model, lifestyle factors like electricity usage, transport habits, recycling practice, and tech consumption have been included for calculating a common Human Sustainability Score. The same chart, like the Eurobarometer reports or World Values Survey reports about different public attitudes and behaviors toward these most important dimensions, is used in the survey. For example, the survey breaks down how much renewable energy is adopted and at what rate does recycling occur, creating input variables in our model, thus underlining the importance of detailing survey source data. Such alignment confirms our feature definition, and augmenting those modeling insights with broad, region-specific data has implications for tailoring accuracy into models with interpretability. Therefore, using such literature, our project aims to convert raw, survey-generated insights to a deep AI-driven sustainability score, which could be useful in policy-making or individual behavioral change promotion.

3 SYSTEM ARCHITECTURE

This is a multi-layered dynamic structure that incorporates individual lifestyle data with macro environment trends for a complete sustainability evaluation: a system architecture where the Human Sustainability Score model is based. It begins with an intuitive User Interface, where users input essential data-frequent electricity consumption per month, modes of transport, recycling practices, buying electronic devices, etc.^[4] It should be remembered that this stage of data collection is vital because it lays down the foundation upon which the model construction takes place.

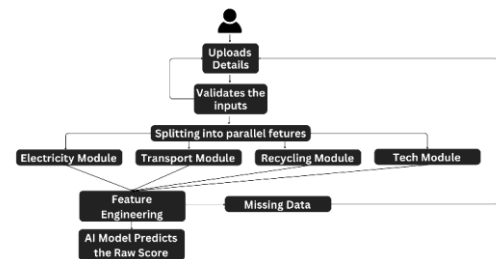


Figure 1: "User Input and Data Flow Diagram"

This representation demonstrates how original data flow from users into the system, which emphasizes the need for correct and full data entry to achieve reliable results. It immediately undergoes strict validation by the system upon submission. Thereafter, it goes through that stage whereby the system checks data for either missing or erroneous inputs and prompts the user to correct them so only high-quality data progresses into processing. After validation, the raw data are routed into special modules designed to deal with different aspects of environmental impact. As an example, the Electricity Module converts kilowatt-hours into CO₂ equivalents considering factors such as renewable energy already utilized (for instance, solar panel integration). The Transport Module will calculate emissions based on a user's choice of travel methods and fuel types. Further environmental footprint warehouse quantification is conducted by recycling and tech

modules, which process recycling percentages and electronics purchase data, respectively.

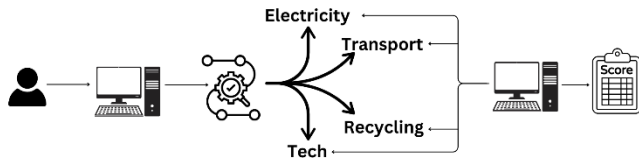
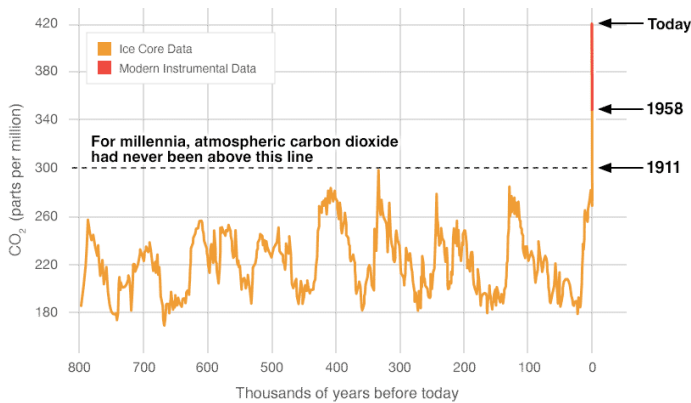


Figure 2: "Modular Data Processing Workflow"

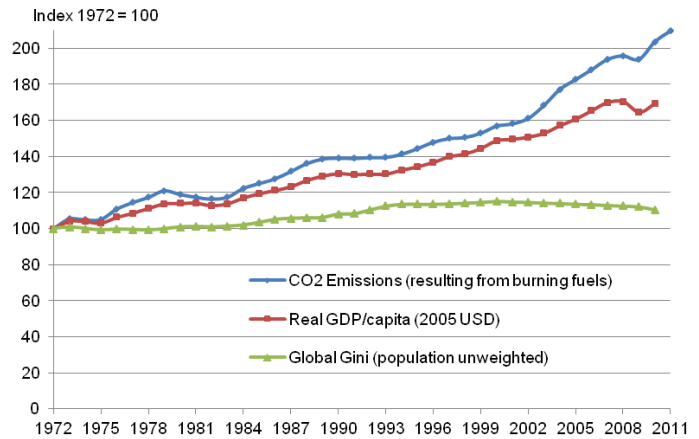
This modular design allows for extreme accuracy since each module is designed to transform an array of inputs to a standardized output that can be measured. The standardized outputs-which are now referred to as engineered features-are fed into our AI Model. This model applies modern machine learning techniques to extract relationships among various sustainability parameters(usually referred to as the input parameters for the model) so as to come up with a raw or primary sustainability score after learning from historical data inputs and trend adjustments. This raw score is then refined by a subsequent scoring engine, which applies rules of weighting for domain meaning. For instance, the engine might increase penalties for activities that disproportionately contribute to CO₂ emissions (high frequency in air travel) and reward those actions that reduce environmental damage (high recycling rates). Therefore, the final output is a holistic sustainability score that renders not only the environmental footprint of an individual but personalized recommendations that drive for future improvement.



Graph 2: "CO₂ Levels Over Time"

To enrich the model further, we incorporate macro-environmental knowledge through two significant graphs. The first is an epidemiological representation of CO₂ levels over time, historical context relative to the present urgency of emission reduction, and a calibration graph for the model toward informing conversion factors and weight adjustments to ensure that individual scores are underpinned by real-world environmental trend evaluation.

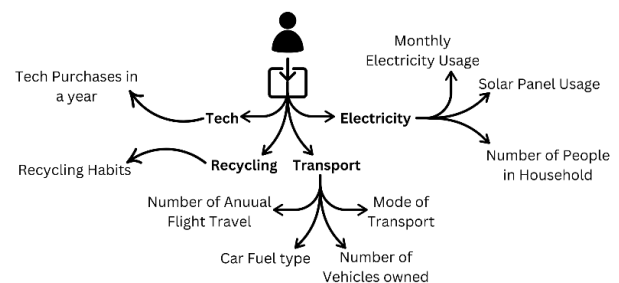
The second graph relates to results in sustainable development, showing the mutual effect of economic growth on emissions with social factors like inequality. The illustration helps not only to contextualize the personal scores but also demonstrates how individual contributions create more significant societal impacts.



Graph 3: "Outcomes of Sustainable Development"

The foundation of strength lies in integrative architecture. It made connected granular user data with global environmental trends and provided accurately actionable, dynamic real-time sustainability score. Another one sharpener of attraction is the feedback loop embedded into the system. The score and recommendations determined for the users can be subsequently used by them to update their data or take actions to correct some facets; the system learns from such updates, thus improving continuously its accuracy and relevance over time.

Based on this we made Input graph and how it is divided into 4 Modules



Graph 4: "User Input Graph"

It illustrates the factors on Input score for Human Sustainability, regarding different aspects of individual life through which they might contribute towards the general sustainability impacts. These factors are categorized into four major parts that include use of electricity, transport impact, recycling habits, and technology purchases. Within each section, it indicates what specific inputs will contribute to calculating the sustainability score of an individual.

"Electricity Usage" refers to energy consumption in a household. This includes Monthly Electricity Usage, which determines the

carbon footprint depending on the energy consumed. Solar Panel usage is also taken into account as it offsets CO₂ emissions using renewable energy. The number of persons in the household also matters, since given household usage, an average per-person figure can be translated.

Whatever form of transport an individual uses would be analyzed in the transport impact section concerning how these types of mobility affect the environment. In this section, the Mode of Transport is perhaps the most significant term, as walking, cycling, and public transport reduce emissions while the private vehicle uses a lot of emissions. The more number of Vehicles Owned, the more Carbon output, owing to emissions from more vehicles. Another significant term is Car Fuel Type -CO₂ emission increases with the use of petrol and diesel cars, while the more environmentally friendly ones are electric or hybrid. In addition to these, Number of Annual Flight Travel is included because air travel adds huge numbers to an individual's emissions.^[5]

Recycling Habits include a multitude of factors that enhance sustainability through resource conservation and waste reduction. The model estimates whether a person recycles, and if so, what percentage of their waste is being recycled. Recycling keeps 0% waste at the landfill and thereby reduces environmental damage, having a favorable impact on the sustainability score.

Last but not least is Tech Purchases, those deals concerning consumer electronics that influence how technology affects the environment.^{[6][7]} Tech Purchases in a Year shows how often individuals get gadgets since acquiring a new electronic product more often means a lot goes into making them and disposing of it subsequently. The smaller this number is, the more appropriate lifestyle becomes sustainable.

In conclusion, therefore, the Human Sustainability Score system does not just act as a mechanical calculator that provides a static figure; it is a progressive, developing, AI-driven instrument that synchronizes individual conducts to global environmental imperatives. Thus, with such designs, users will possess not just a number but much more significant contextual views on how their daily decisions affect a larger ecological picture. By embedding macro-level trends within personalized data of the user, our model brings a different standard of sustainability assessment-scientific and practically useful innovations for driving sustainable change.

DATASET REQUIREMENT

A comprehensive and well-organization of the dataset is one of the prime tasks for constructing a good and accurate AI model for predicting the Human Sustainability Score. The dataset will have to cover a diverse array of individual lifestyle factors that affect environmental sustainability. Major focus areas are energy consumption, transportation habits, recycling behavior, and technology usage. In energy consumption, this dataset should reflect the monthly household electricity usages of the individual household in kilowatt-hours (kWh) and whether any of that consumption is offset by renewable sources, such as by solar panel usage—say, indicated as a percentage. This is essential because the kWh to CO₂ emissions (using a factor like 632.41 g CO₂ per

kWh) conversion is the backbone of the Electricity Module of the model.

Likewise, in transportation, the dataset should consider the number of vehicles owned by the household and travel behavior, including information on daily commute and annual flight travels, either expressed in kilometers or as a count of flights. This will allow the model to compute emissions associated with transportation, which could be further mellowed with information regarding fuel types of vehicles (petrol, diesel, electric, hybrid, etc.).

Another critical point that should be surveyed is recycling behavior of a household as a percentage of waste recycled. This information gives positive actions toward environmental benefits that have a reducing effect on net emissions. Moreover, data capture of technology consumption takes into account the number of tech or electronic purchases per year, having a negative environmental effect through production and waste generation, where frequent purchases contribute.

Other demographic and contextual variables such as household size and the city or region being lived in further help to refine the model by enabling per capita adjustments and region-specific calibrations. If pre-existing datasets do not combine all these features into one source, one course of action is to merge data collected from several trustworthy sources—down to energy usage data provided by government reports, transportation statistics, municipal recycling data, and market research on tech consumption. In rare cases, where such merging is simply impossible, another option is to apply survey-based collection procedures targeted at capturing such variables directly from individuals. This ensures that the final dataset will be representative as well as granular enough steering a data-driven, adaptive sustainability model.

Feature Name	Data Type	Description	Source/Notes
Electricity Usage (kWh)	Numeric (Integer)	Monthly household electricity consumption in kilowatt-hours.	Utility bills, energy surveys
Solar Panel Usage (%)	Numeric (Float)	Percentage of electricity consumption provided by solar panels (if applicable).	Renewable energy records, surveys
Household Size	Numeric (Integer)	Number of people living in the household.	Census data, survey responses
Number of Vehicles Owned	Numeric (Integer)	Total number of vehicles owned by the household.	DMV records, household surveys
Car Fuel Type	Categorical	Type of fuel used by the vehicles (e.g., Petrol, Diesel, Electric, Hybrid).	Vehicle registration data, surveys
Flight Travel	Numeric (Integer/Float)	Total annual flight travel (in kilometers) or number of flights taken per year.	Airline data, travel surveys
Recycling Habits (%)	Numeric (Float)	Percentage of total household waste that is recycled.	Municipal waste reports, surveys
Tech Purchases per Year	Numeric (Integer)	Number of new technology/electronic items purchased annually by the household.	Market research, consumer surveys
City/Region	Categorical	Geographic location of the household.	Public records, census data
Final Sustainability Score	Numeric (Float)	The target variable computed from the aggregated input factors using the model's formula.	Derived using the defined calculation

Table 1: "Dataset Requirement and its Type"

Already Existing Datasets and Their Approaches

At this point, there is no existing database that collects all of these features in together. Pictures, videos, and other forms of data

have been displayed by a variety of organisations and governmental agencies, but invariably as single entities.

How best to address:

1. Data Merging:

Integrate data sets from a variety of primary reputable sources e.g. utility company Energy Consumption Data, transport Ministry source, recycling statistics from municipal records and consumer electronics purchase data from market research.

2. Survey-based data collection:

Research development and conduct surveys to collect extensive lifestyle information on respondents. This outfitting allows for consolidation of all the required features into one dataset.

3. Synthetic data generation:

In case of real-world data collection being too unrealistic, employ discipline and statistics for synthetic generation using realistic distributions per attribute. This should work with care and against known benchmarks.

Combining these complementary strategies has the potential to achieve a comprehensive and intricate dataset supportive of developing an AI-enabled Human Sustainability Score model.

4 SYSTEM ANALYSIS

MODULE 1- ELECTRICITY

Monthly Electricity Usage: ' E '

Solar Panel Usage Percentage: ' S '%

Number of People in Household: ' H '

1. Raw Calculation:

- Convert to CO₂ emission using:

$$CO_{2raw} = E \times 0.63241^{[12]} \text{ (in kg)}$$

2. Solar Adjustment:

- If the user has solar panels, assume a solar offset percentage S %.

$$E_{eff} = E \times (1 - (S/100))$$

- Recalculate effective emissions:

$$CO_{2eff} = E_{eff} \times 0.63241$$

3. Household Adjustment:

- Divide by household size H to get per capita emissions

$$CO_{2pc} = CO_{2pc}/H$$

4. Non-Linear Scaling:

$$Score_{elec} = 10 \times (1 - (1/(1 + \exp(-k_e(\theta_e - CO_{2pc}))))$$

- The constant k_e adjusts the steepness (e.g., $k_e=0.03$).
- This function produces a score near 10 for very low emissions and near 0 when emissions far exceed the threshold.

MODULE 2-TRANSPORT

Mode of Transport: ' T_d '

Number of Vehicles: ' V '

Fuel Type: ' F '

Annual Flight Travel: ' D '

1. Daily and Annual Emissions:

- Let T_d be the daily transport-related CO₂ emission (e.g., 9 kg for a car).^[13]

$$T_{base} = T_d \times 365$$

2. Vehicle Multiplication:

- If the user owns V vehicles, multiply:

$$T_{vehicle} = T_{base} \times V$$

3. Fuel Type Adjustment:

- If the car fuel is non-electric, add a penalty. For instance, for every liter of fuel consumed, add 2.3 g CO₂ (converted to kg)^[14] as an extra factor. Let F be the penalty factor (scaled appropriately), so:

$$T_{fuel} = T_{vehicle} + F \text{ (if non-electric)}$$

4. Flight Travel:

- With an emission factor of 0.2 kg CO₂/km^[9],

$$T_{flight} = D \times 0.2$$

5. Total Transport Emission:

$$T_{total} = T_{fuel} + T_{flight}$$

6. Scaling with Logistic Function:

$$Score_{trans} = 10 \times (1 - (1/(1 + \exp(-k_t(\theta_t - T_{total}))))$$

- A suitable constant might be $k_t=0.001$
- Define a transport threshold θ_t (e.g., 5000 kg CO₂/year).

MODULE 3-RECYCLING

Recycling Percentage: 'R %'

- Direct Scaling with Diminishing Returns:

$$\text{Score}_{\text{recy}} = 10 \times (1 + \exp(-k_r(R - \theta_r)))$$

- Use a modified logistic function (or a piecewise linear function) to account for diminishing returns as R approaches 100%.
- Here, θ_r could be set to 50 (target recycling rate), and k_r might be around 0.1.
- If recycling is "No," then set $\text{Score}_{\text{recy}} = 0$.

MODULE 3-TECH

Number of tech/electronics purchases per year.: 'N'

- Non-Linear Penalty:

$$\text{Score}_{\text{tech}} = 10 \times (1 + \exp(-k_{\text{tech}}(N - \theta_{\text{tech}})))$$

- Use a logistic decay function to represent increasing negative impact with each additional purchase:
- For instance, set $\theta_{\text{tech}} = 2$ (the desired purchase level) and $k_{\text{tech}} = 1$ to reflect rapid decay in score beyond that point.

FINAL SCORE

$$\text{Final Score} = (\text{Score}_{\text{elec}} + \text{Score}_{\text{trans}} + \text{Score}_{\text{recy}} + \text{Score}_{\text{tech}}) / 4$$

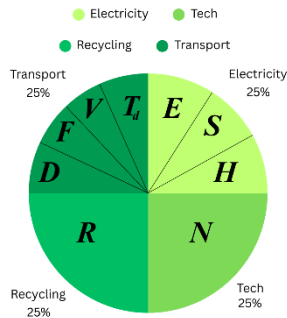


Figure 3: "Pie – Chart Illustration of Inputs Variables"

• SAMPLE WORK:

It includes a sample input table based upon 15 hypothetical cases where each row has a scenario based on a user. In the case, we take key input variables as monthly electrical use (in kWh), use of solar panels as a percentage, household size, number of owned vehicles, flight distance travelled in a year (in km), recycling rate (in percentage), and the number of tech purchases made in a year. Ultimately, each of them is processed in one of the four modules: Electricity, Transport, Recycling, and Tech, and subsequently works on individual scoring. For example, in case 1, the user is

consuming 300 kWh in a month with no solar offset, and they're residing in a household of three. Such an input would yield a very high Electricity score (9.84 out of 10) against the per capita emissions below a specified threshold.

Transport is another module that provides emissions information from the daily commute, number of vehicles, and flight travel. For instance, in this case, as he has one vehicle and no flight travel, it has calculated his transport score as 8.47. Recycling will convert that reported 50% recycling into a score of 5.00-determined via a logistic function that considers 50% as the median level. Meanwhile, the Tech module will also regard that two tech purchases carry a score of 5.00. The individual scores from the modules are then summed and given equal weights to give a final score of 7.08 to Case 1.

It follows that within the table, the lesser electricity consumption levels, together with higher recycling rates or fewer tech purchases made, usually translate into higher scoring against those modules. So for example, Case 5, which has very low electricity consumption of only 100 kWh, has no transport (zero vehicles and flight), and has a very high recycling rate (90%), scores 9.60 overall in a lifestyle believed to be very sustainable. On the other hand, Case 14, with an Electricity score of 3.45, is probably undercut by his high per capita emissions and low recycling, leading to a final score of 5.10. The table speaks volumes of the model because here is where one can see how several lifestyle factors will interact through nonlinear calculation (with the use of logistic functions) to coalesce into module scores. Moreover, it portrays the effects of individual parameters on the overall score, thus making comparisons among different scenarios. With these sample inputs and scores in mind, one can gauge how changes in specific areas such as electricity efficiency or recycling practices may foster valuable improvements translating into higher final Human Sustainability Scores.

Case	Electricity (kWh)	Solar (%)	Household	Vehicles (V)	Flight (km, D)	Recycling (%)	Tech Purchases (N)	Electric Score	Transport Score	Recycle Score	Tech Score	Final Score
1	300	0	3	1	0	50	2	9.84	8.47	5.00	5.00	7.08
2	500	0	4	1	1000	80	3	9.74	8.20	9.53	2.69	7.54
3	200	50	2	1	0	30	1	9.94	8.47	1.19	7.31	6.73
4	600	20	4	2	0	60	4	9.77	1.72	7.31	1.19	5.25
5	100	0	1	0 (none)	0	90	0	9.84	9.93	9.82	8.81	9.60
6	800	0	5	1	500	40	3	9.52	8.34	2.69	2.69	5.81
7	450	30	2	1	200	70	1	9.53	8.42	8.81	7.31	8.52
8	700	10	3	1	1500	55	2	8.82	8.04	6.22	5.00	7.02
9	350	40	2	1	0	20	2	9.82	8.47	0.47	5.00	5.94
10	250	0	2	1	500	80	1	9.74	8.34	9.53	7.31	8.73
11	550	25	3	1	300	65	3	9.67	8.40	8.17	2.69	7.23
12	400	10	2	1	0	50	2	9.30	8.47	5.00	5.00	6.94
13	900	15	4	1	800	75	4	9.15	8.26	9.24	1.19	6.96
14	350	0	1	1	0	30	1	3.45	8.47	1.19	7.31	5.11
15	650	40	3	1	250	85	2	9.72	8.41	9.70	5.00	8.21

Table 2: "Sample User Input Results"

5 COMPARISON OF RESULTS

As comparison table, this outlines the key features of existing sustainability tools to the proposed model, particularly data sources, applicability, entry into AI, format for output or score, use difficulties, paywall status, and overall availability.^[11] Both the WWF and CarbonFootprint.com depend on self-user reports through online calculators, thus giving access to the whole world, but their computation of results tends to be a rather static rather non-AI approach. Oroeco and My Carbon Action work with self-reports combined with some levels of machine learning (the check mark in the AI-Based column indicates this) for their insights into lifestyle patterns. However, there are also some limitations regarding the global applicability and ease of use of the tools. On the other hand, it has an edge over the other because it takes in multi-factor input into a unified AI-powered system, that is to say, covering electricity, transport, recycling, and tech habits. Therefore, it would include complex analytics and global applicability to offer a more nuanced sustainability score-without-boll, thus making it an accessible and comprehensive view of personal environmental impact.

Reference	Sources	Globally Available	AI-Based	Output / Score	Difficulty	Paywall	Publicly Available
WWF	User self-reports (Online Calculator)	✓	✗	✗	✓	✓	✓
CarbonFootprint.com	User self-reports (Web-based)	✓	✗	✗	✓	✓	✓
Oroeco	Lifestyle data, spending patterns	✗	✓	✗	✗	✓	✓
My Carbon Action	User surveys, consumption logs	✓	✓	✗	✗	✓	✓
Proposed Model	Multi-factor inputs (electricity, transport, recycling, tech, etc.)	✓	✓	✓	✗	✗	✗

Table 3: "Comparison with Various Pre-Available Models"

Yayzy is an online sustainability tool that calculates one's carbon footprint and gives users an overview of the environmental impact. Primarily, it uses a fixed and predetermined set of formulae and input variables to produce quick results. In contrast, the Human Sustainability Score model adopts a multi-layered AI-powered approach that considers a very broad set of lifestyle variables into one score, from electricity consumption (factoring for solar panels), to detailed transportation metrics (including vehicle ownership, fuel type, and flight travel), recycling behavior, and tech purchases. Yayzy relies more on static calculations; our model, on the other hand, utilizes logistic scaling functions within modules so that non-linear alterations mirroring the true cumulative impact of personal actions can be accounted. Furthermore, our system is set with the intention of a multi-AI modal system, where the user can choose among the modes for different analytical routes depending on the richness and type of available data. Besides that, our model is envisioned to consider some macro-environmental factors like region-specific data and district-wise GDP, which will provide a contextual layer from which both personal and policy interventions can be informed. Thus, in all, Yayzy is a very handy and useful baseline assessment, while our approach is adaptive, modular, and data-rich, allowing for a more subtle, scalable, and actionable sustainability

assessment.

Aspect	Yayzy	Our Model	Advantage of Our Approach
Calculation Methodology	Uses fixed formulas and predefined averages	Uses modular, AI-driven approach with logistic scaling functions for non-linear adjustments	More accurately reflects cumulative, individual-level impacts
Input Granularity	Limited set of inputs (typically energy, transport, basic recycling info)	Captures multiple, detailed lifestyle factors (electricity with solar adjustment, vehicle details, flight travel, recycling, tech)	Offers a comprehensive view of diverse sustainability factors
Adaptability & Flexibility	Static, one-size-fits-all model	Multi-AI modal system allowing users to choose different models based on available data and desired level of detail	Provides tailored insights and can evolve with new data
Contextual Integration	Generally does not integrate regional or socio-economic data	Plans to incorporate city/district-level data and GDP to contextualize sustainability scores	Supports targeted policy recommendations and localized sustainability strategies
User Engagement	Quick, accessible, but limited in actionable insights	Designed to deliver personalized, data-driven feedback that guides behavior change	Empowers users with clear, actionable insights and encourages sustainable practices
Scalability	Suitable for general use with minimal customization	Built as a scalable framework that can be refined and expanded to include additional lifestyle parameters and economic indicators	Enables continuous improvement and adaptability to different regions and datasets

Table 4: "Yayzy vs Our Model"

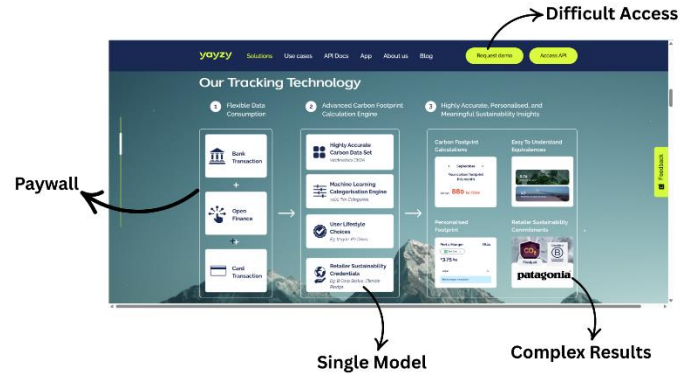
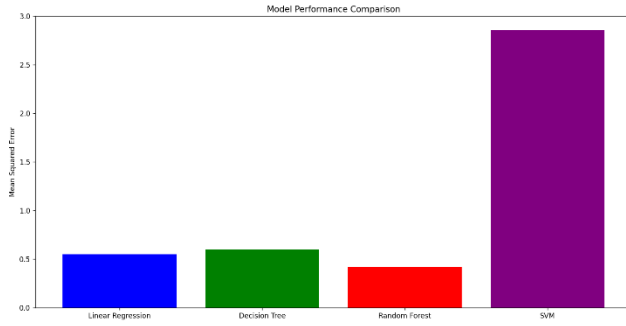


Figure 4: "Yayzy UI Complexity"

These days, it is considered standard practice, and often required by demand, to have tools with sustainability assessments paid with money. Most of these platforms are subscription-based or charge for licensing fees, as they have involved a huge investment in data, advanced analytical methods, and continuous refinement of models. The costs of procuring high-quality datasets from diverse sources, developing trusted AI algorithms, and ensuring compliance with regulations are heavy. Oftentimes, hence, the market has shifted to paid products, which in turn ensure the sustainability of a service provider who may also offer insights that tend to be more accurate, customizable, and up to date. Users can take advantage of features, such as real-time on-the-ground data integration, regional analytics, and personalized recommendations, that usually come as add-ons to the free versions. Further, such monetization would ensure the possibility of continuous research and development for further improvements and innovation in the fields of sustainability assessment.

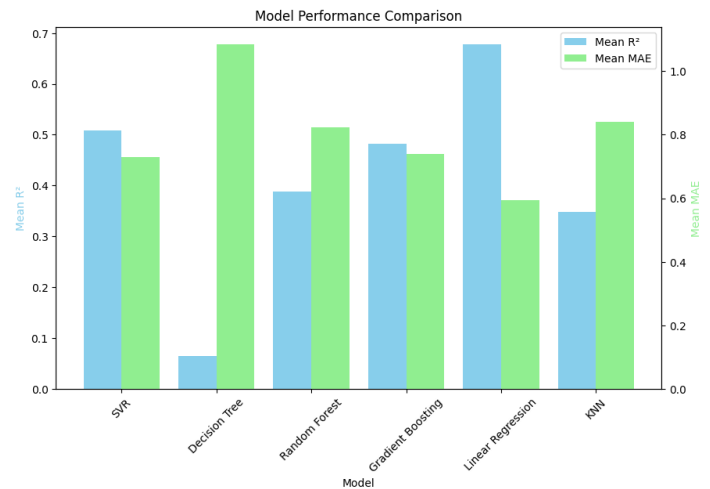
PERFORMANCE AND RESULTS



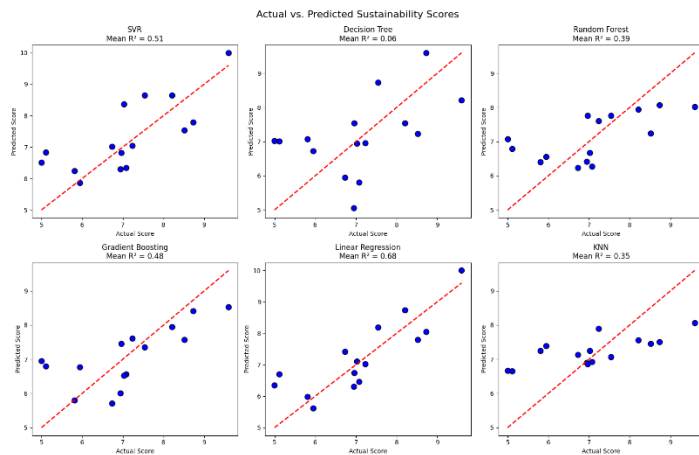
Graph 5: "Model Performance Comparison"

The bar chart illustrates how the four machine learning models - Linear Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) - are evaluated in terms of the Mean Squared Error (MSE). The less the MSE, the better is the accurate prediction. Between all models, Random Forest has the lowest MSE, therefore, it is considered the best model in this comparison. Compared with errors of Linear Regression and Decision Tree, Decision Tree performs slightly worse. SVM has the highest MSE by a wide margin, indicating that it is poor performance on this dataset, which is likely to be due to improper hyperparameter tuning, nature of the data, or SVM not being well suited for this specific problem. All in all, the results impelled that Random Forest is the most suitable model for this dataset while SVM needs pieces of further improvement or reconsideration.

performance is the Mean R^2 , indicated within each subplot. This value ranges from 0 to 1 and measures goodness of fit; the higher the value, the better the predictive accuracy. The exceptional model evidenced here by the highest R^2 of linear regression, 0.68, suggests that this model is the most competent to make this prediction. The model of the decision tree, conversely, has produced an R^2 of 0.06, indicating a very poor fit for this application. SVR and Gradient Boosting performed moderately well, attaining values of 0.51 and 0.48 R^2 , which may reflect reasonable but not outstanding predictive capability. Random Forest and KNN exhibit a moderate predictive power of 0.39 and 0.35 R^2 , respectively, just below that of SVR and Gradient Boosting, albeit with the same predictive worth. The scatter plots back this numerical exposition, with linear regression showing the best fit of observation points to the diagonal line while Decision Trees dispersed most widely. Overall, this image gives a clear overview focused on comparing the models' predictive accuracy and underlining the importance of model choice on the basis of performance metrics and visual fit.



Graph 7: "Multi Model Performance Comparison"

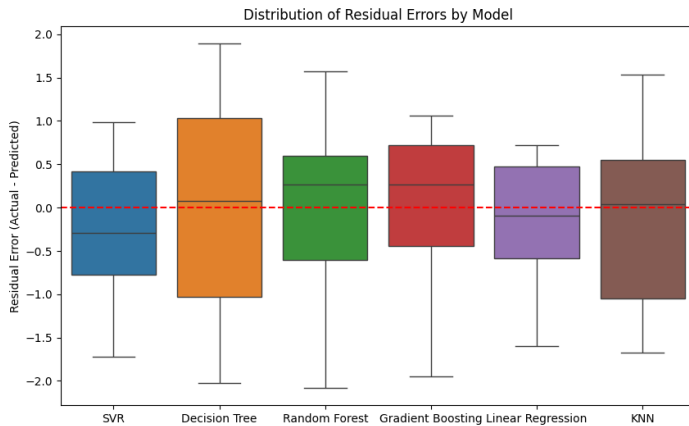


Graph 6: "Multi Model Score Comparison"

In this image, six distinct machine learning models—Support Vector Regression (SVR), Decision Tree, Random Forest, Gradient Boosting, Linear Regression, and K-Nearest Neighbors (KNN)—are judged against each other on predicting "Sustainability Scores." The visual consists of a 2x3 grid with each subplot representing one model. Every subplot features a scatter plot with the actual sustainability score on the x-axis and the corresponding model's predicted score on the y-axis. A red dashed diagonal line is drawn across all subplots, serving as a criterion for perfect prediction where both actual and predicted scores are in perfect concordance. The primary measurement of model

This image shows a comparative study of six machine-learning models: Support Vector Regression (SVR), Decision Tree, Random Forest, Gradient Boosting, Linear Regression, and K-Nearest Neighbors (KNN) with respect to mean R^2 and mean absolute error or mean MAE. The grouped bar chart representation has two adjacent bars, one for Mean R^2 (light blue) and the other for Mean MAE (light green), each for a model. The left y-axis describes the Mean R^2 ranging from 0 to 0.7, while the right y-axis describes the Mean MAE ranging from 0 to 1. The model names can be read on x-axis placed at 45-degree angle for the ease of reading. The title given to this chart is "Comparison in model performance," which adequately summarizes its use. Notably, Linear regression performs best with the highest Mean R^2 value of about 0.68 coupled with low Mean MAE (around 0.37), showing more accurate prediction with a minor error amount. In fact, concerning Mean R^2 (close to 0) as well as Mean MAE (near 1.1), the Decision Tree model shows the poorest performance. On the other hand, models like SVR, Random Forest, Gradient Boosting, and KNN exhibit moderate performance, with Mean R^2 values

lying between the ranges of 0.35-0.51 and Mean MAE values between 0.45-0.8. This chart vividly represents trade-offs of effective predictive ability versus errors for various models and makes Linear Regression the most effective model for this particular task.

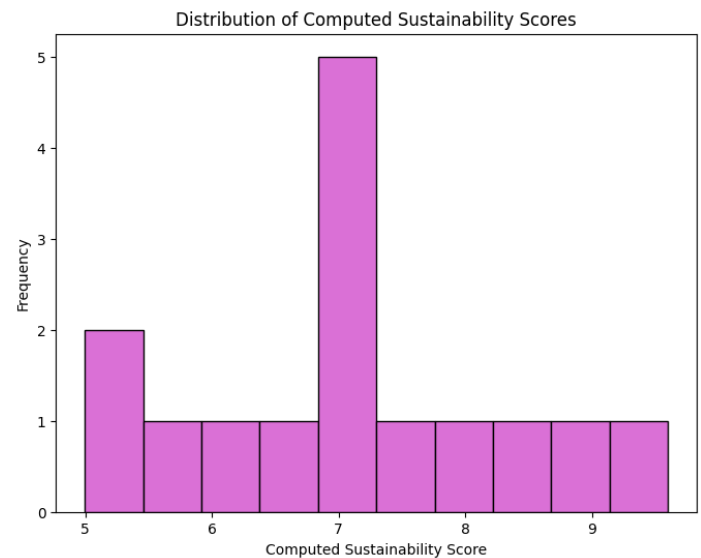


Graph 8: "Multi Model Distribution Graph for Residual Errors"

This image shows a comparative analysis of the distribution of residual errors by six machine learning models: Support Vector Regression (SVR), Decision Tree, Random Forest, Gradient Boosting, Linear Regression, and K-nearest neighbors (KNN). All models will be exercised based on how each one is illustrated by the boxplot. Each of them forms a box in the image, which shows the model's distribution of residual errors, which is equal to the discrepancy between the actual and predicted values. The y-axis explains how this discrepancy varies from about -2 to 2, while the x-axis is made up of the names of the different models. A red dashed horizontal line is found at $y=0$ to show that it is the level under consideration with regard to error value. The title "Distribution of Errors for Each Model" makes it clear what the information in this chart will cover. Boxplots then show the distribution, median, as well as the extremes of residual error values by each model. It can be inferred that Linear Regression and Gradient Boosting have the least breadth in their boxes which indicate lower variance in their errors and for that reason, considered as more consistent. On the contrary, the Decision Tree model presents a much wider range and a higher median error, exposing lesser accuracy and more variance. Finally, moderate spreads and medians are presented for SVR, Random Forest, and KNN, revealing their spreading tendency. The positioning of those boxes against the zero-error line, however, indicates the tendency that each model has of over or under prediction. In general, this figure helps give an idea of the errors distributions for each model and gives an overview of their respective consistencies and accuracies in prediction.

This image shows a comparative description of the distribution of residuals across six machine learning models, namely, Support Vector Regression (SVR), Decision Tree, Random Forest, Gradient Boosting, Linear Regression, and K-Nearest Neighbors (KNN). Visualization is conducted in a boxplot in which each

model is represented by a box displaying the distribution of its residual errors, calculated as actual values minus predicted values. Residual error is plotted on the y-axis from -2 to +2, while model names line the x-axis. A reference point for zero error is shown with a red dashed horizontal line at $y=0$. The title "Distribution of Residual Errors by Model" makes it evident what the chart is about. A box plot shows the availability, median and possible outliers of the residual error for each model. It clearly specifies that Linear Regression and Gradient Boosting are the models that indicate the least level of breadth in their boxes, which would mean variations are significantly lower in their errors, and thus they may be more consistent. The accuracy of the Decision Tree is lower and varies greatly since its range and median error is much wider. SSVD, RF, and KNN showed mild spreads and medians. The position of the boxes against the zero-error line indicates the tendency of the models to over or under-predicted. It gives insight into the distribution of errors per model and provides a general picture of their consistencies and accuracies in failure.



Graph 9: "Score Distribution of Our Data"

The histogram illustrates the distribution of Computed Sustainability Scores. The title of the chart makes clear its aim: Distribution of Computed Sustainability Scores. The x-axis is labeled Computed Sustainability Score, with a range stretching from 5 to 10 along the computed score line. The y-axis has the label of Frequency, indicating how many times a score appeared in the data set. The bins of the histogram are divided into 10 spread-out sustainability scores. All bars are light orchid-colored, with an outer black border for easy identification of the bins. The data highlight clustering around a score that returned a greatest frequency of 5. Frequencies then continue to decline on either side of 7, indicating that scores near 7 were more frequently observed in the dataset. Scores from 5 to 6 and from 9 to 10 hold least frequencies, indicating that such extreme values were rare. In all, the histogram has done a great job visually analyzing the distribution of computed sustainability scores, showing central tendency with some spread of the data.

It provides a holistic assessment about varying performances of machine learning models in predicting "Sustainability Scores" together with a distributional overview of the scores themselves. The first scatter plot indicates that Linear Regression predictions are indeed the best because of the high R^2 and prediction actual score corresponds closely, while Decision Tree model prediction is the least effective. The bar plot showing Mean R^2 and Mean MAE strengthened this finding and indicated superior performance by Linear Regression and the failure of the decision tree. Next, the box plots showing residual errors or deviations depicted the consistency and variance in prediction associated with these models and brought out relative stability of errors in predictions by Linear Regression and Gradient Boosting as models. However, the last histogram showed computations from each of the computed sustainability scores, revealing convergence of scores around 7 to indicate central tendency. All in all, these images comprehensively evaluate model performance whilst providing a better understanding of predicted variable distribution, iterating the significance of model selection and learning out of prediction error and data distributions.

IMPACT

Most importantly, it can be a personalized, measurable number for people's footprints, thereby having a strong potential positive impact on them. By providing people with a direct correlation score for lifestyle choices-such as energy consumption, transportation practices, recycling, and tech purchases-they will explain how their choices directly affect the environment. This opens up visibility into the meaning of sustainability; instead of abstract worries about climate change, the figures will shed light on moving toward behaviors in which recycling or energy savings can be perceived directly in score improvement. Thereby, over time, such changes could result in decreased CO₂ emissions through more effective resource utilization and greater involvement and adherence to sustainability principles. Such scores when aggregated and analyzed will also provide better information for planning community initiatives and public policies toward green infrastructure and public programs. In short, it is knowledge and gives personal motivation, thereby sending ripples that will certainly also benefit the larger society and environment.

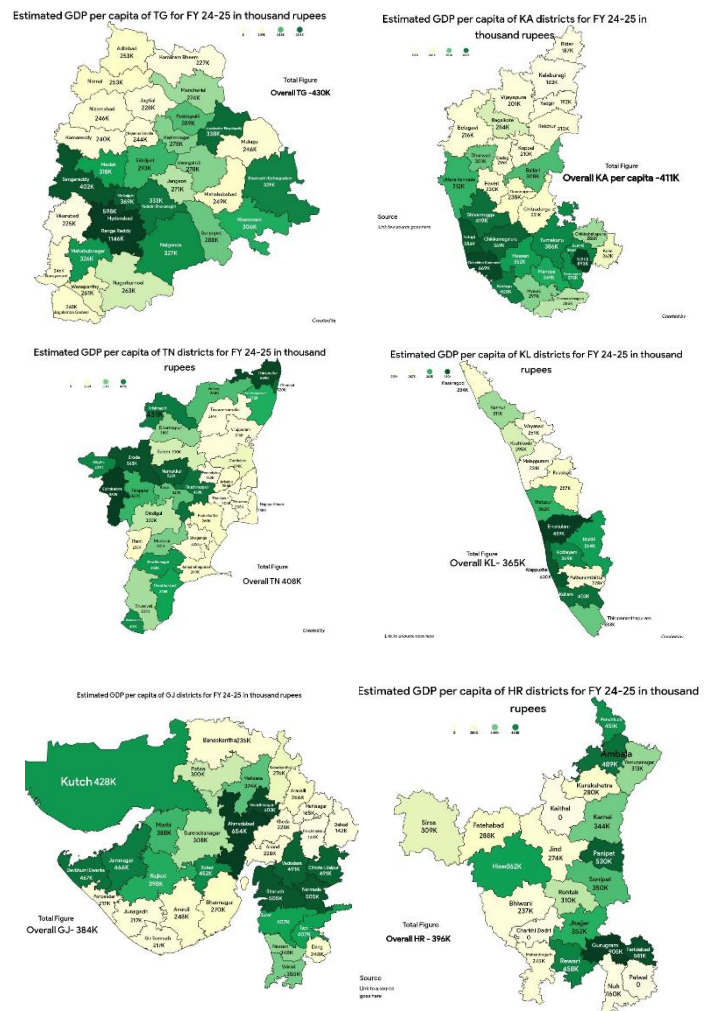
6 CONCLUSION

Summarizing sufficiently, the Human Sustainability Score model presents an innovative and more holistic approach to assessing individualized environmental impacts. This model takes into account major lifestyle influences such as electricity use, transport, recycling, and technology purchasing, aggregating them into one all-encompassing measure. With unique integration and AI methods, through non-linear scaling logistic functions, the model normalizes differing data inputs while accurately reflecting the contribution of each component to overall sustainability profile.

This framework can help individuals in improving their environmental footprint, while also providing policymakers and researchers with a concrete tool to evaluate and promote sustainable behaviors. Out of its modular nature, a potential for arbitrary improvement arises, therefore allowing the model to maintain relevance across regions and time as new information flows in. In the horizon's future, evidence-oriented and worthwhile approaches to sustainability will become more feasible.

FUTURE SCOPE

In further iterations, the model can be improved by incorporating geospatial factors for sustainability evaluation at a regional and urban scale. Local sustainability scores can thus be derived from the AI model by anchoring data to specific cities and districts, making possible comparisons across regions.



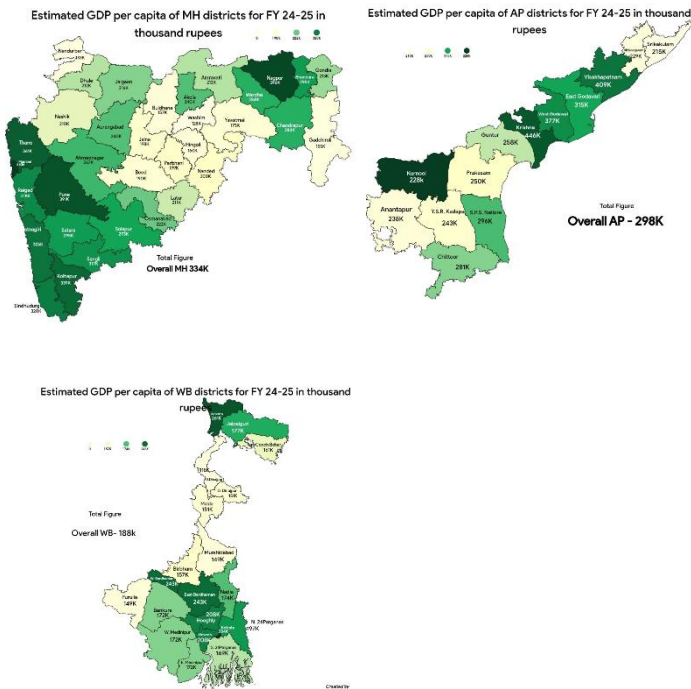


Figure 5-13: "District -Wise GDP of Telegana-Karnataka-TamilNadu-Kerala-Gujarat-Haryana-Maharstra-Andhra Pradesh-West Bengal"

For instance, by integrating GDP and other city-level information for the districts from 9 states, one can assess the linkages between environmental performance and economic activities. This could establish trends, such as whether those areas with relatively higher GDP sustain higher sustainability scores because of more extensive investments in green technologies and infrastructure, or conversely, in some areas, economic growth is coupled with increased consumption of resources and pollution of the environment.

In addition to that, when geographical data are introduced, the factors affecting energy infrastructure, transportation, and waste management will therefore incorporate region-specific considerations. A rather more complex view necessary for policymakers and urban planners emerges, which offers focused insights into where and which areas are driving and where sustainability is being lagged. For example, an area with higher GDP could also have access to renewable energy, efficient public transport, and recycling programs, which would all contribute to a higher sustainability score. Alternatively, regions with rapid economic growth would struggle to practice sustainability if their investments in green infrastructure failed to match their expansion. Combining sustainability scores and GDP data on a district level would then reveal such causative factors so decision-makers could formulate appropriate designs for interventions for promoting environmental stewardship across varying urban contexts.

An advantage for future research could also lie in the potential for advanced geospatial objectives- out-coupled with GIS- to visualize sustainability scores alongside any economic correlates on interactive maps. With such an integration, clients could perform spatial analyses, such as clustering regions with similar attributes or monitoring changes along time. All in all, geo-referencing city and district data against other economic indicators provides a complete frame of reference for understanding the interaction between environmental sustainability and economic development, which in the end will give rise to informed and practical strategies for our greener future.

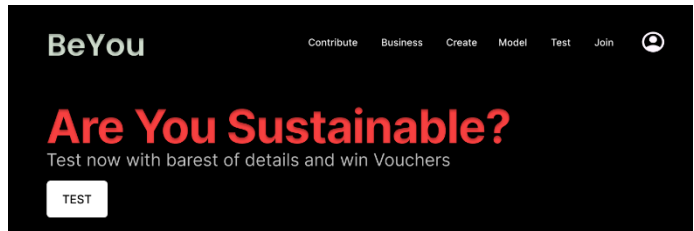


Figure 14: "Aim UI of Our Website"

The future development of the model will be as part of a multi-AI modal, which will integrate lifestyle data for sustainability assessment and provide users an option based on different AI models, input configurations, and dataset types. This way, it will give choice for selecting an analysis mode suitable for the data availability or particular needs of a user. For example, a user with just basic entry data like electricity and transport usage could apply lightweight modeling that will employ a reduced set of features while those with a more extensive dataset (e.g. recycling habits, tech purchases, even regional socio-economic indicators) could opt for a more demanding multiple factor AI model.

This flexible multi-modal system will be modular in architecture and dynamic in its selection and integration of several AI algorithms-from traditional regression techniques to advanced ensemble and deep learning models. The users will not only be able to compare the result of the different AI algorithms, such as SVM, Decision Trees, Random Forests, Gradient Boosting, etc., on the same platform, but this approach is expected to improve user engagement along with the cross-validation of results, thereby making sustainability scores further robust and transparent.

In addition to that, the framework would be extensible at accommodating a variety of datasets. It would also include pre-processing pipelines that would automatically adapt to the different data styles and quality, thus keeping the model adaptable, whether data is based on surveys or public records or sensor-based. The goal of providing such improved flexibility and customization would be to make the multi-modal system operate in a customized accord with the individual user condition to promote informed decision-making and ultimately either in personal or community behavior towards sustainability.

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